

MicNest: Long-Range Instant Acoustic Localization of Drones in Precise Landing

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Flying Meals



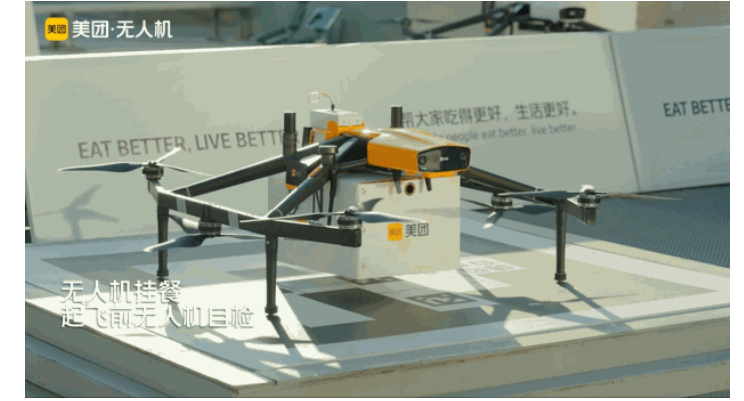
Main Procedures of Drone Delivery



1. Online Order



2. Package



3. Take off



4. Fly to the destination



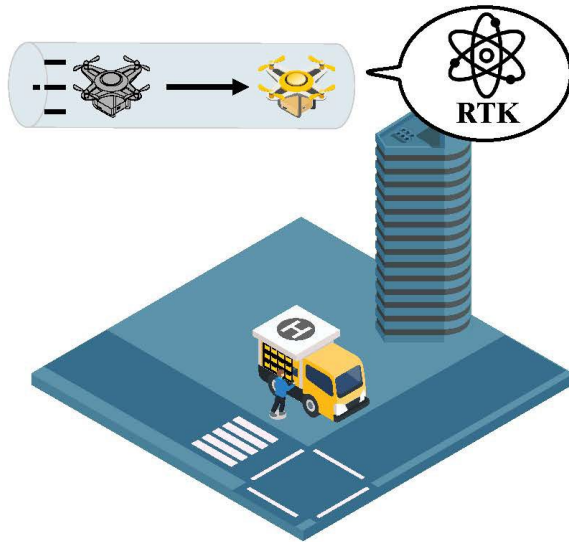
5. Land and dock onto the self-collection station



6. Fetch

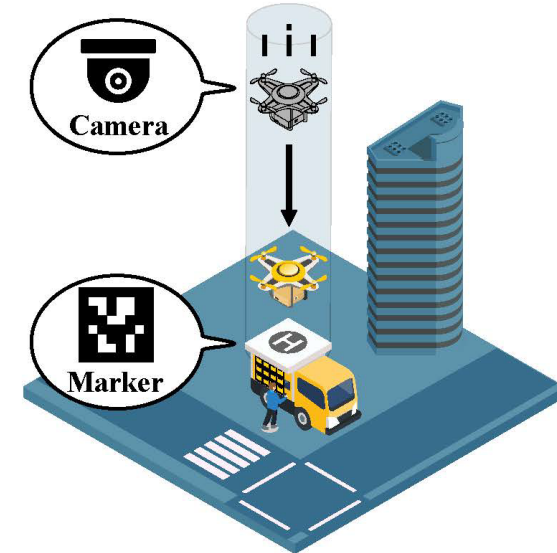
State-of-the-Art in Precise Landing

Step 1: Horizontal approach



- ✓ High Precision
- ✓ Vast Coverage
- ✗ Unstable Performance

Step 2: Vertical descent

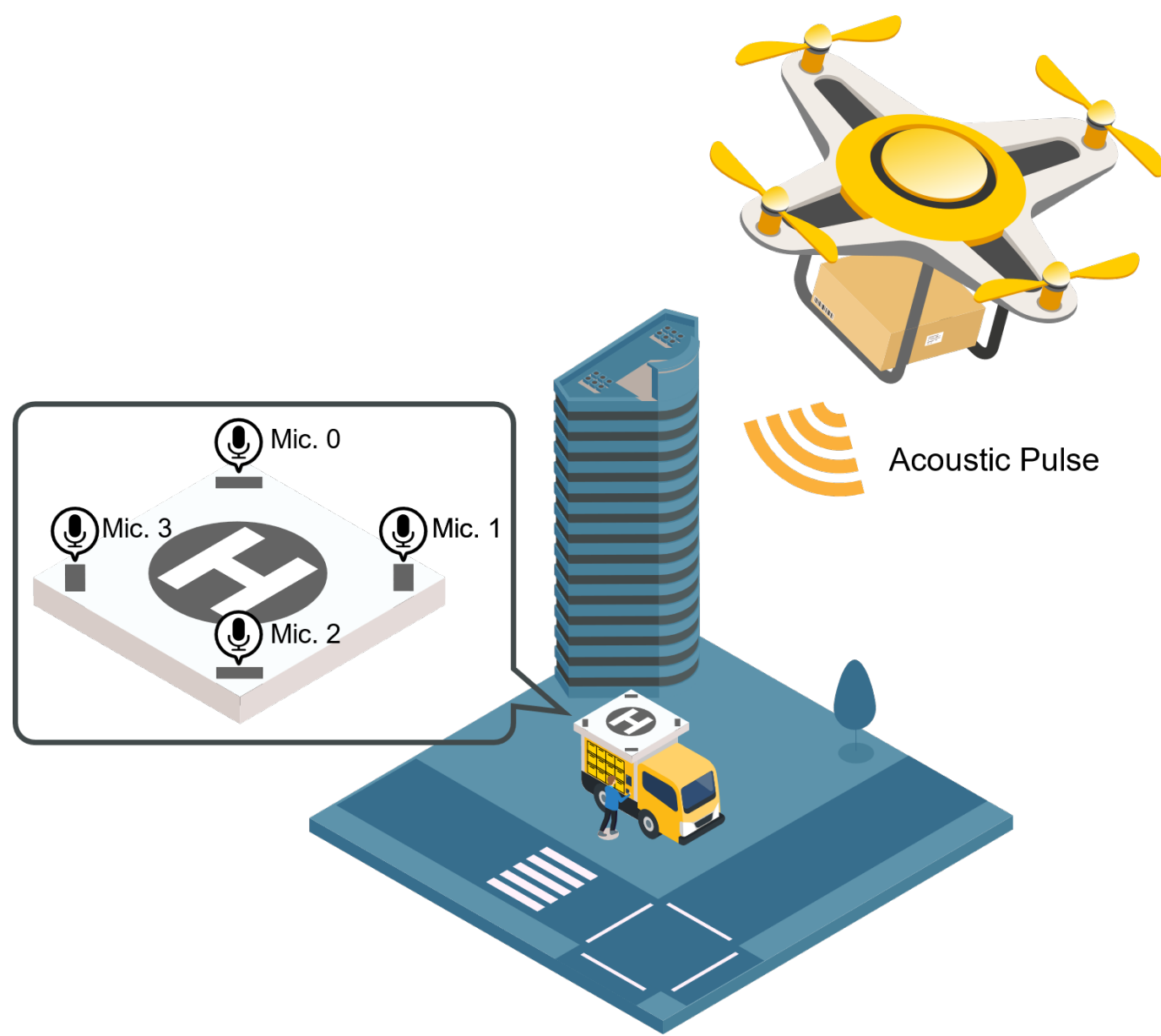


- ✓ High Precision
- ✗ Limited Coverage
- ✗ Sensitive to lighting conditions

Problem: Existing SOTA techniques can not provide reliable localization services for precise landing.

MicNest:

Acoustic Localization System for Drone Landing

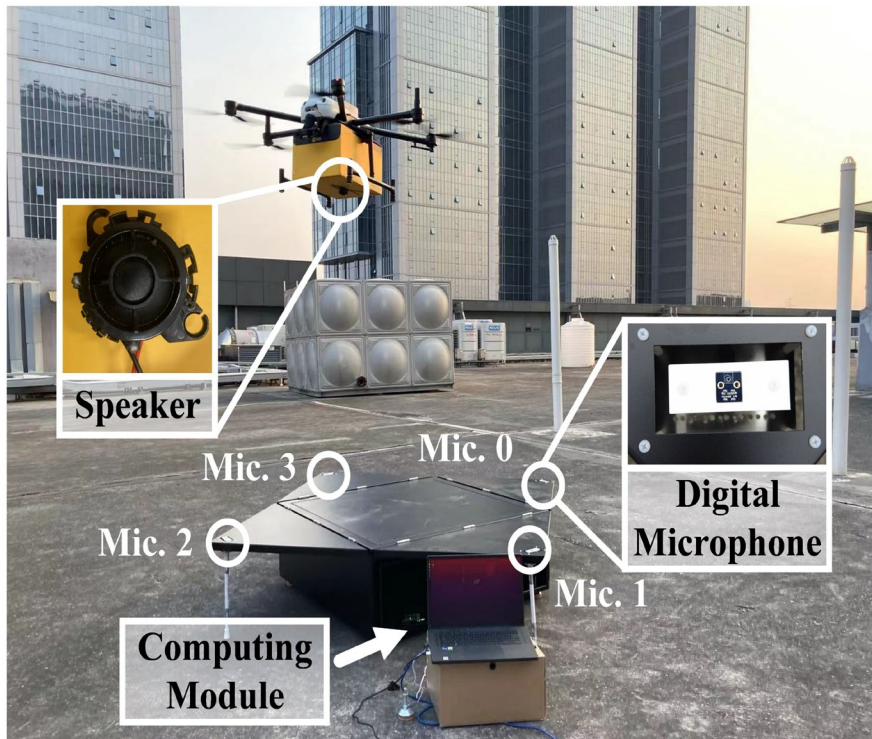


120 m
Operation Range

0.53%
Relative Localization Error

20 Hz
Location Update Rate

MicNest in a Nutshell



Hardware

Pulse Transmission



The pulse transmitted by each drone:

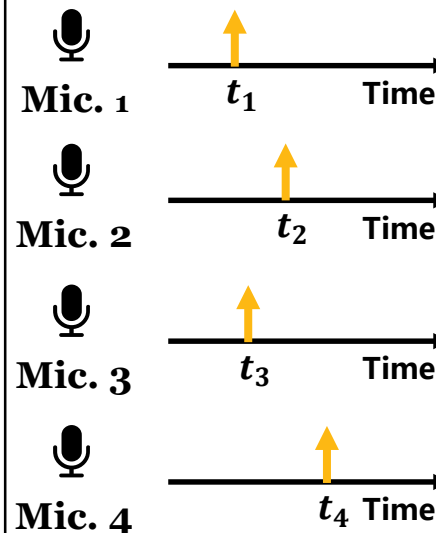
$$\mathbf{s} = [s_0, s_1, \dots, s_n, \dots, s_{N-1}]^T$$

Each code s_n is an independent random gaussian variable

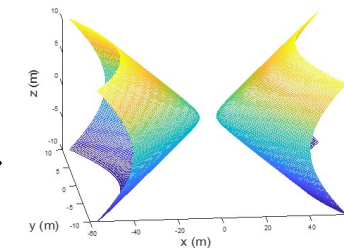
Advantages:

1. Comfortable to hear
2. Concurrent Localization of multiple drones (CDMA)
3. Resistance to spoofing attack

Pulse Detection



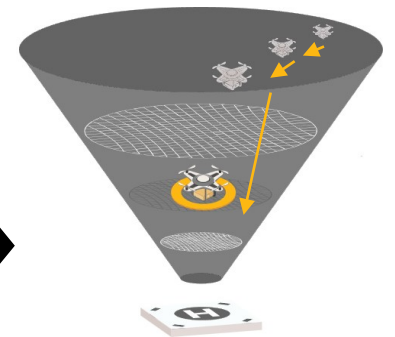
Hyperboloids



$$abs(|PM_1| - |PM_3|) = 2a_1$$

$$abs(|PM_2| - |PM_4|) = 2a_2$$

Localization



By exploiting the onboard altitude sensors, the drone can thus be localized.

Workflow

Why Acoustics?

Spatial Resolution

- The **spatial resolution** R (absolute ranging resolution) of wireless signal is proportional to **signal speed** c , and is inversely proportional to **signal bandwidth** B :

$$R \propto \frac{c}{B}$$

- Due to **low sound speed**, the acoustic signal can provide pretty promising spatial resolution.



UWB: Decawave DWM1000

- Bandwidth: 500 MHz – 1 GHz
- Spatial Resolution: **>10 cm**



FMCW: TI IWR1443

- Bandwidth: 500 MHz – 4 GHz
- Spatial Resolution: **>3.75 cm**



Acoustics

- Bandwidth: 24 KHz
- Spatial Resolution: **0.7 cm**

Challenges

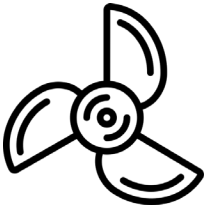
Low SNR



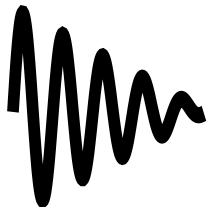
Limited
Tx Power



High
Noise Floor



Strong
Interference



Serious
Attenuation

Non-linear Distortion

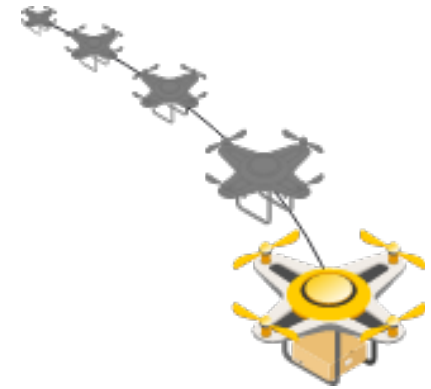
- Low sound speed leads to significant Doppler distortion.

$$\mu \propto \frac{v}{c}$$

- The flight control loops rapidly change the drone velocity, resulting in non-linear distortions.



Low Latency



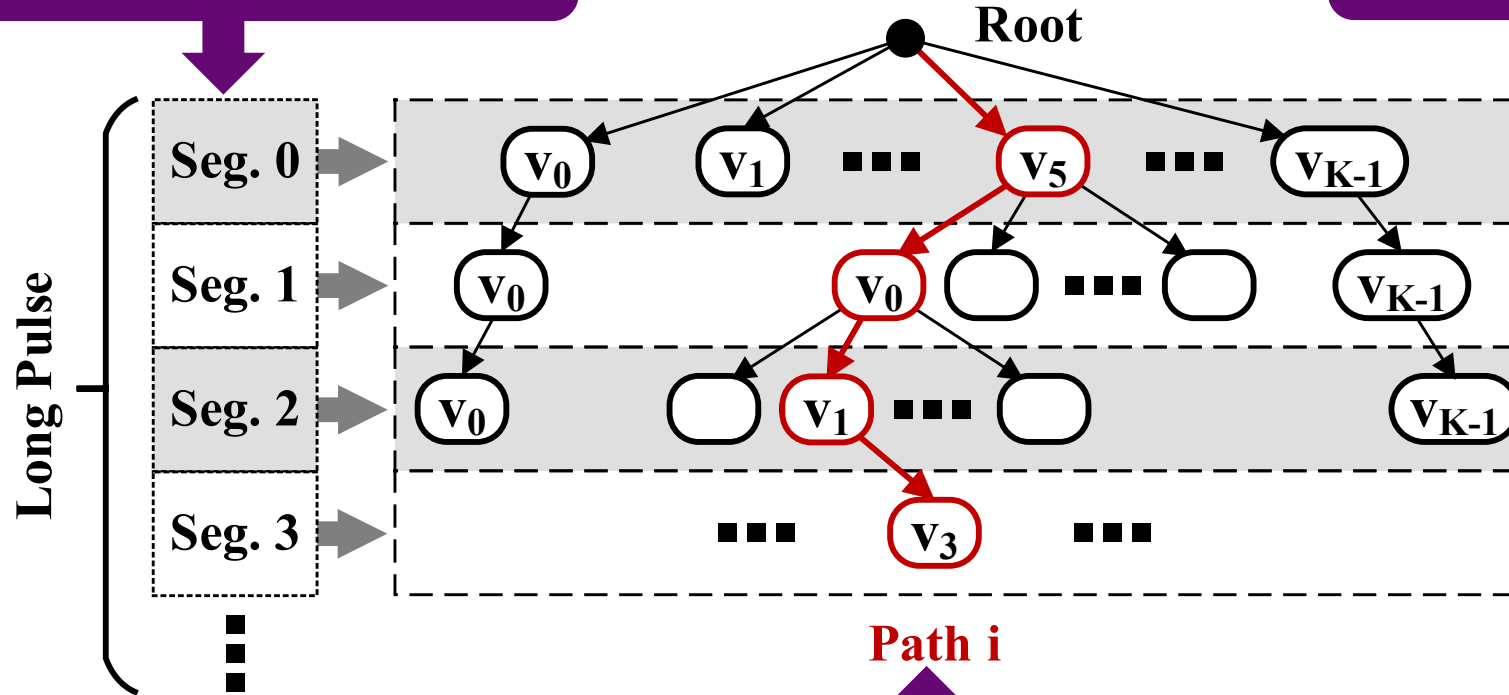
- MicNest is used to navigate moving drones.
- Signal processing must withstand the latency constraint imposed by the nature of flight control loops.

How to detect **low-SNR** pulses with **non-linear distortion** efficiently

Matched Filter Tree (MFT)

cut one pulse into
multiple short segments

nodes at each level correspond
to the possible drone velocities

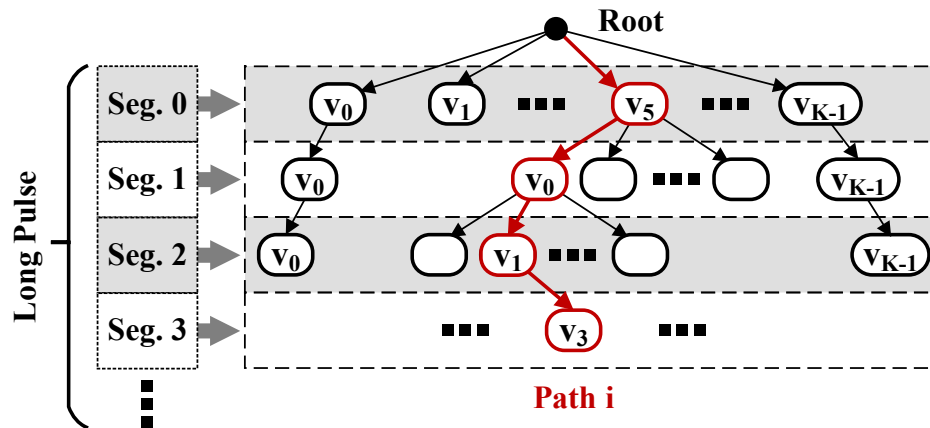


nodes along a tree path represent a
set of compensation coefficients

Key Idea: convert pulse detection to a tree search problem

Matched Filter Tree (MFT)

Goal: search for the tree path that can minimize the non-linear distortion



Exhaustive Search

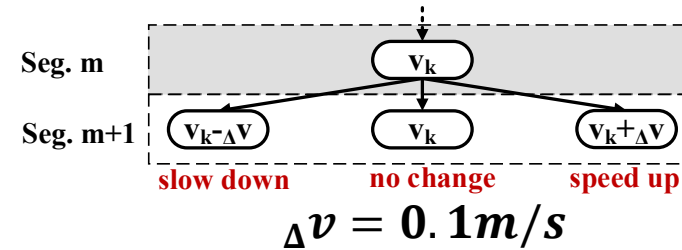
- For each tree path, perform the following operations:
 - Compensate** segments by using coefficients
 - Concatenate** compensated segments into a new pulse template
 - Correlate** the new template with the received signal.
- After visiting all paths, select the tree path with the maximum correlation value;

- The search space grows **exponentially** with #segments
- Each path involves an **expensive** correlation operation

Accelerate Tree Search

Tree Pruning

- The drone velocity does not change abruptly
- The branching factor of the tree can be **reduce to 3**

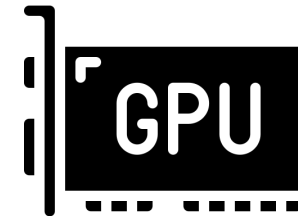


reduce search space



Correlation Acceleration

- The pulse-audio correlation can be decomposed into multiple segment-audio correlations.
- This computing scheme (**vector adding**) can be highly parallelized by GPU.

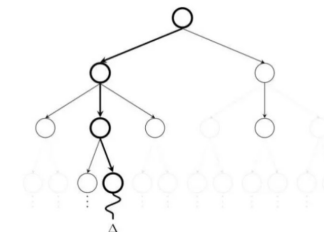


reduce time cost of search path



Heuristic Search

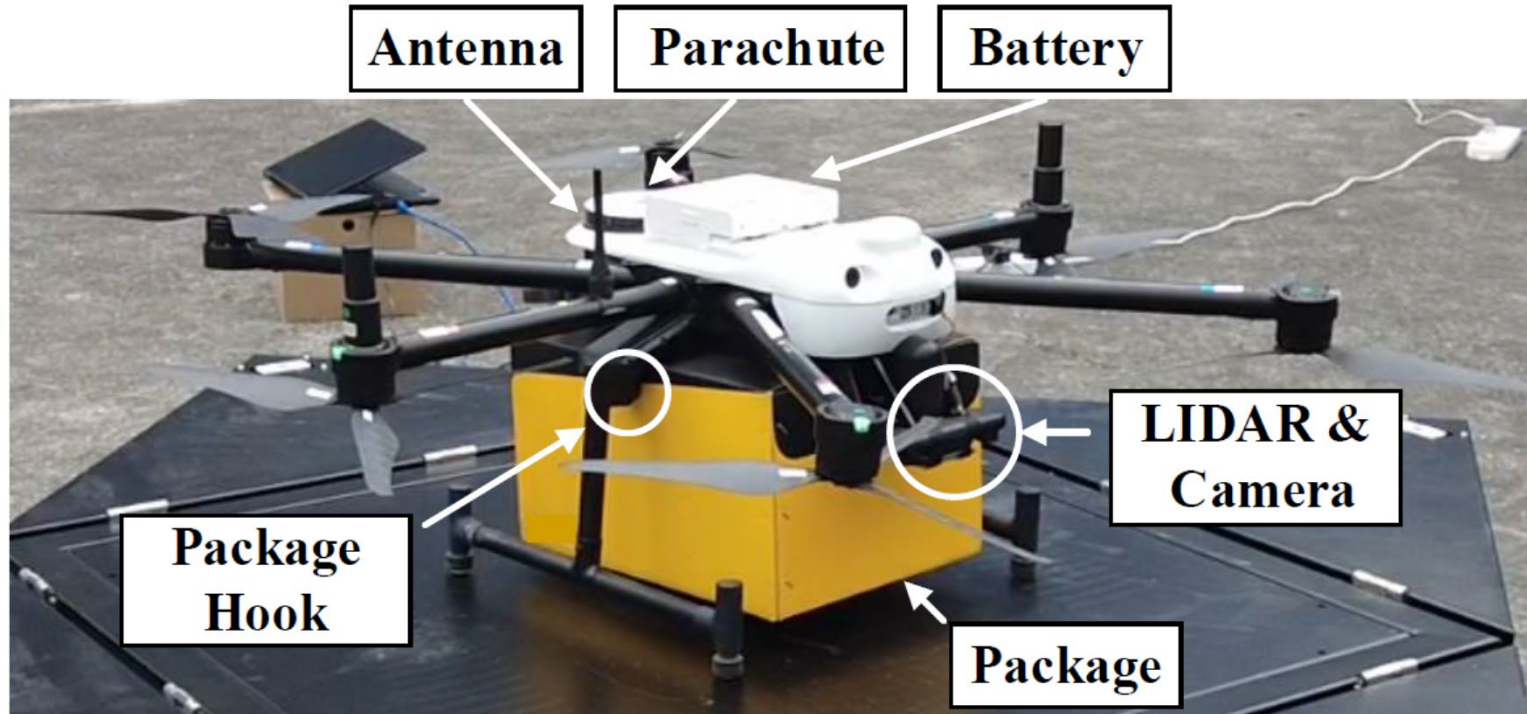
- This idea is similar to **Monte-Carlo Tree Search (MCTS)**
- We can exploit visiting history to quickly dive towards the solution.



reduce total visit count

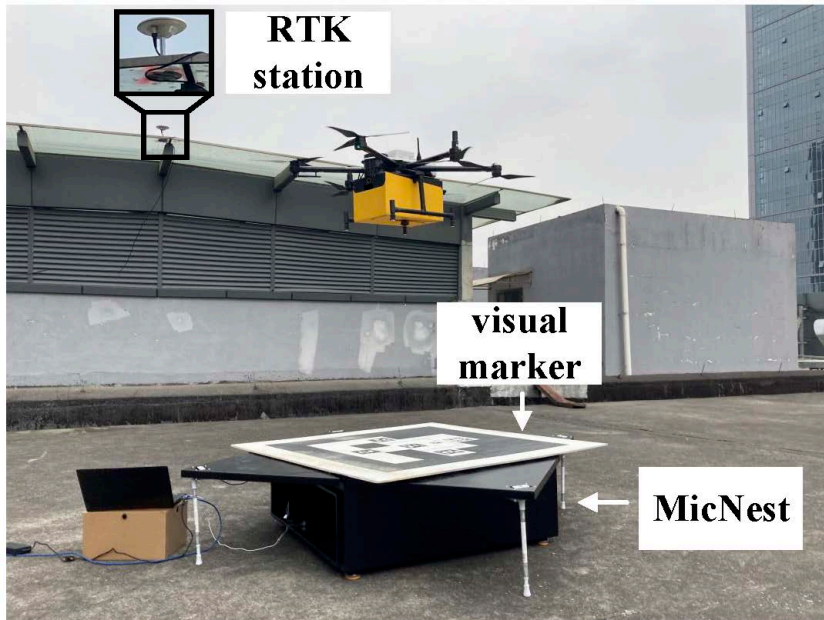


Delivery Drone



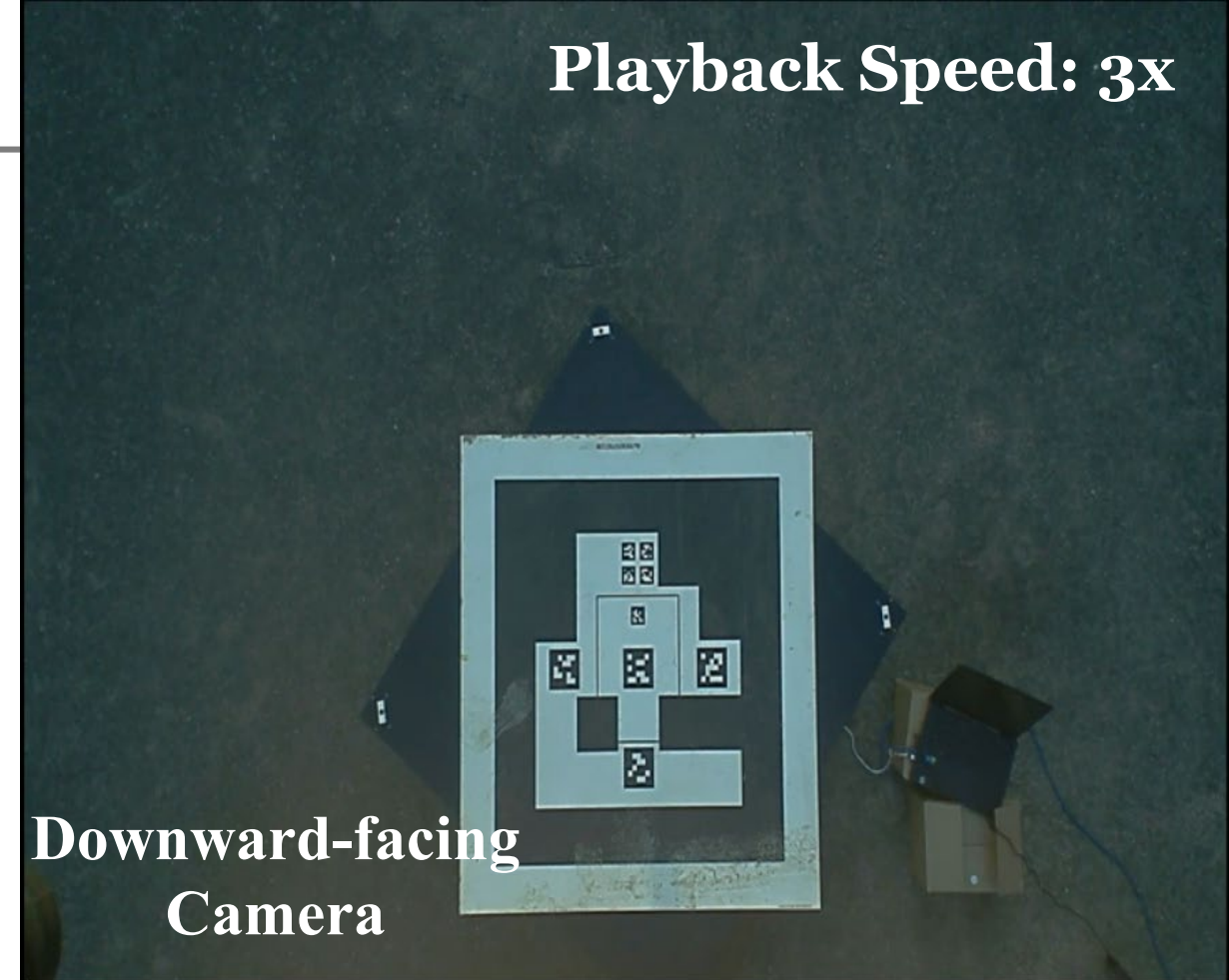
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Evaluation

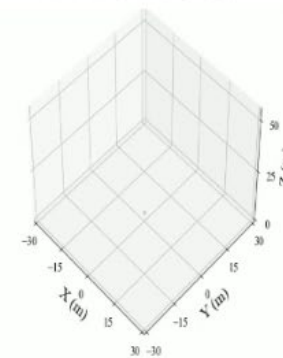


Baselines:

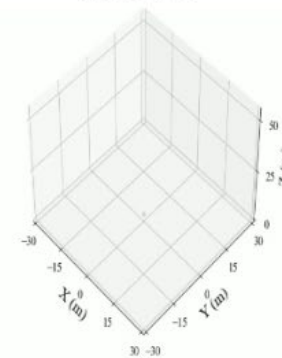
- RTK
- Visual Marker



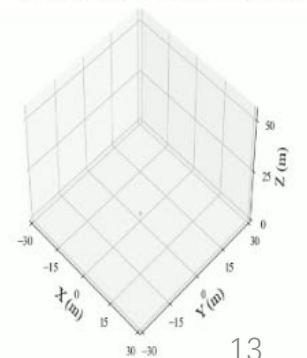
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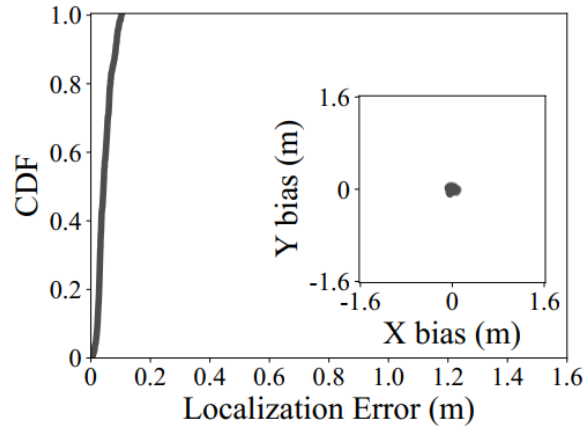
RTK



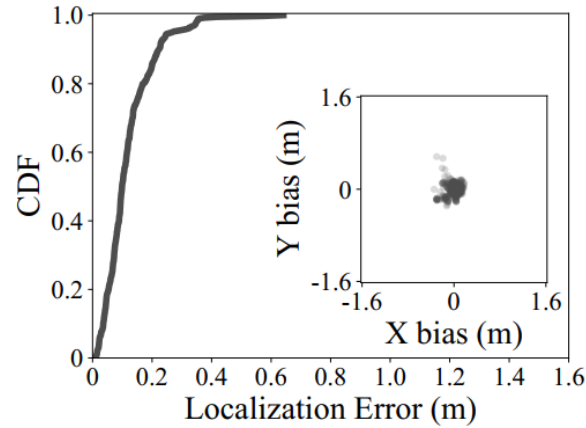
Visual Marker



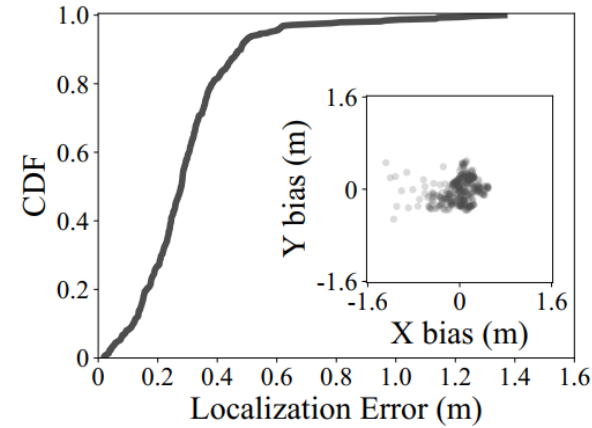
Impact of Altitude



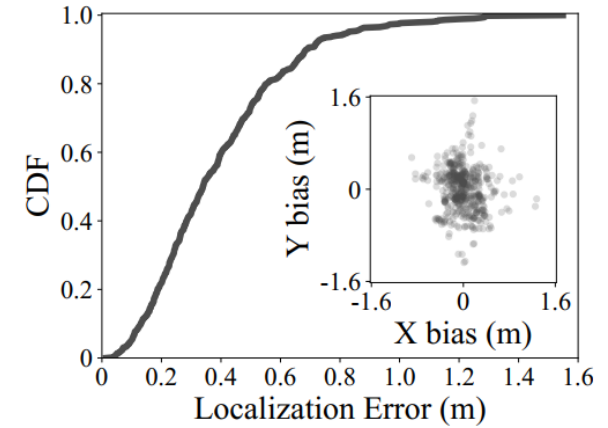
0m-20m altitude



20m-50m altitude



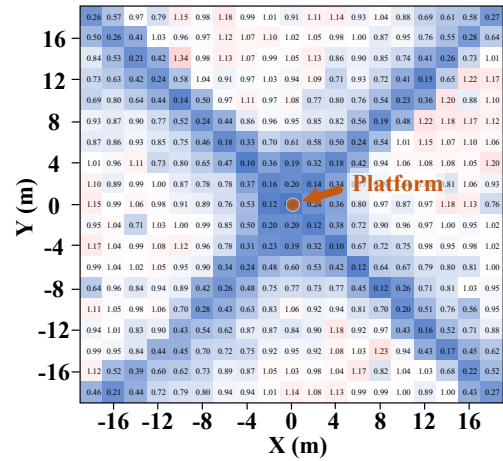
50m-80m altitude



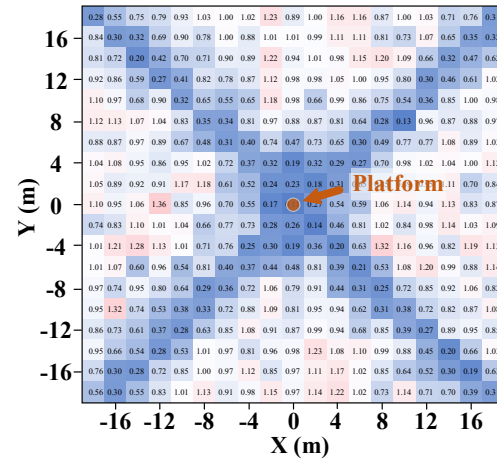
80m-120m altitude

Feature 1: localization error decreases as the altitude decreases

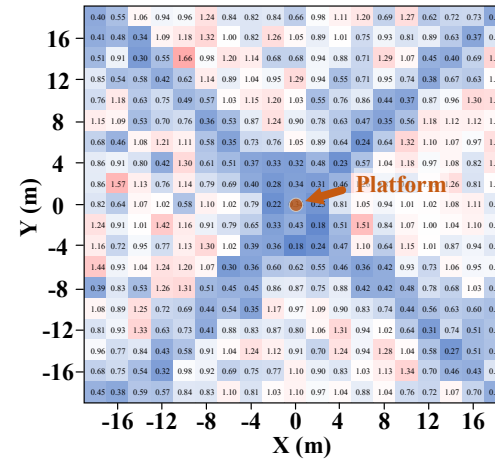
Impact of Horizontal Distance



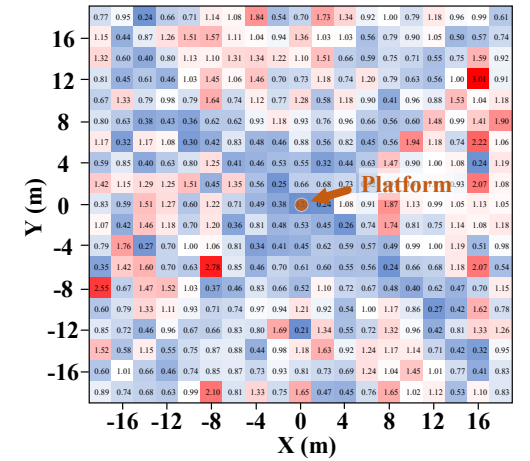
altitude 25m



altitude 50m



altitude 75m



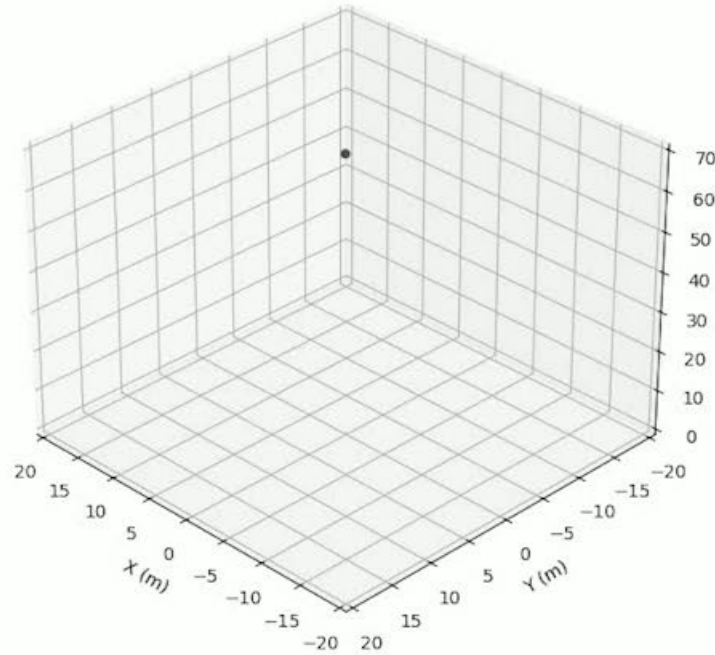
altitude 100m

Feature 2: Localization error decreases as the drone approaches the platform

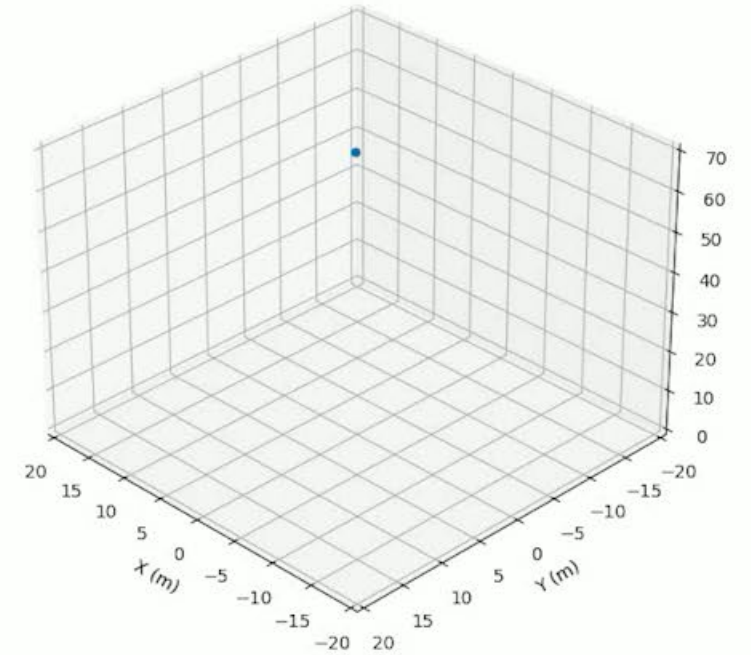
Navigating Drone Landing



MicNest

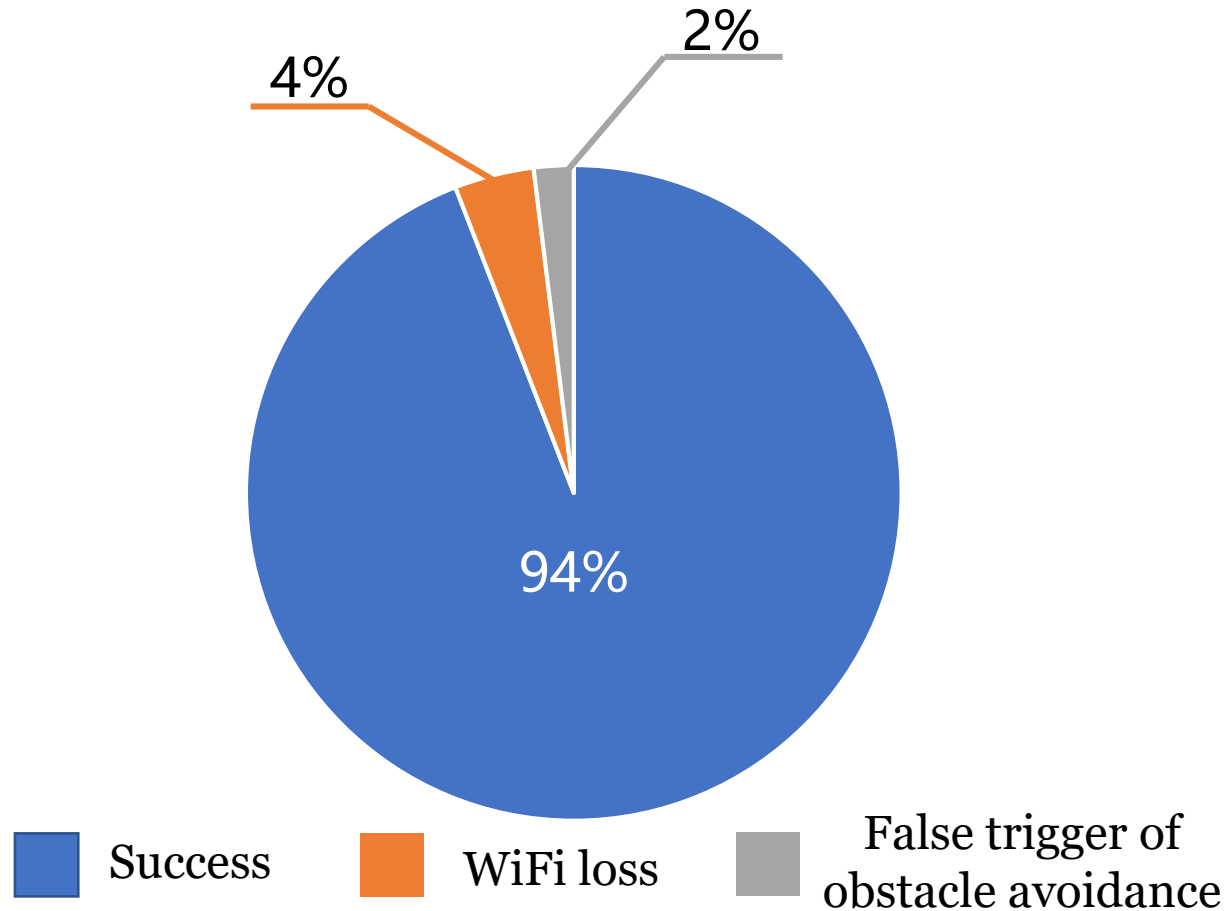


RTK

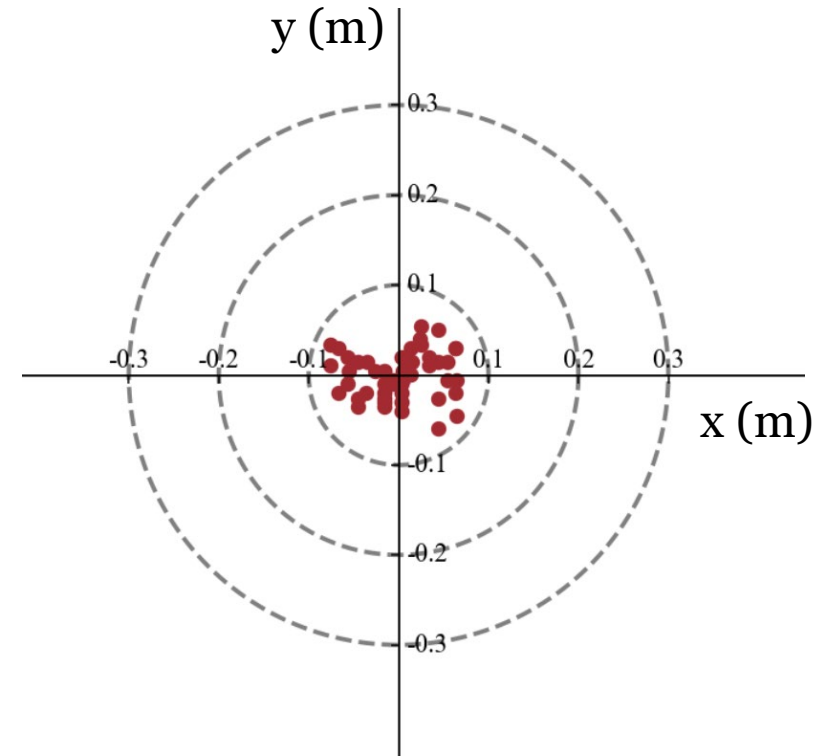


Playback Speed: 3x

Navigating Drone Landing



Statistics (50 continuous flights)



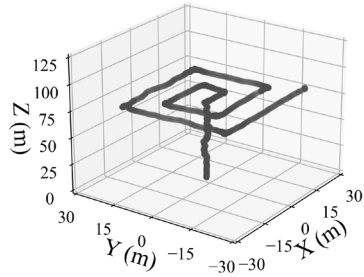
Distribution of landing points

MicNest can navigate drone landing **robustly and accurately**

Summary

- **MicNest** enables precise landing of drones using acoustic signals.
- The key enabling technologies we present are **MFT**, a novel pulse detector that models the problem as a tree search problem
- MicNest is able to localize a drone **120 m** away with **0.53%** relative localization error at **20 Hz** location update frequency.

Milestones and Future Work



**Real time
localization**

500

**Accumulated
test flights**



**Successful
navigating landing**



**Incorporating
MicNest ...**

2021.10

2021.12

2022.01

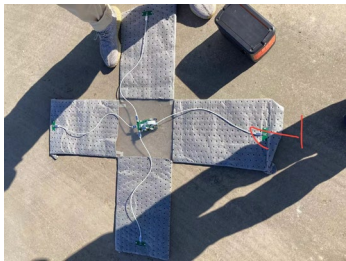
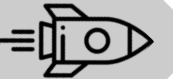
2022.07

2022.07

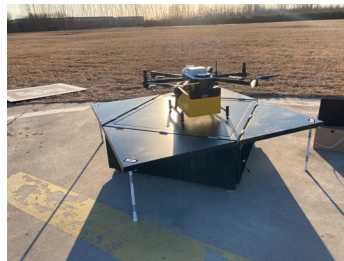
2022.08

2022.09

2022.10
(ongoing)



**Initial
microphone array**



**New
microphone array**



**Approved for use
in flight control
systems**

100

**Accumulated
navigations**

Thank You!



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